

High Flow Nasal Cannula(HFNC) Initiation Algorithm

Indications	1. Mild –moderate respiratory distress 2. Anticipated Intolerance to mask or interface Common diseases for HFNC use ☐ Bronchiolitis most common indication ☐ Other respiratory diseases causing respiratory distress such as pneumonia, asthma
Contra-indications	☐ Significant respiratory distress(relative) / respiratory failure ☐ Hemodynamic instability ☐ Acute respiratory Distress Syndrome with significant hypoxemia ☐ Encephalopathy



Settings	Infant : Start Flow at 2l/kg/min, Fio2 60% Child : Start Flow 1 l/kg/min , Fio2 60% ☐ Aim for saturation 93%-94% ☐ Titrate Fio2 to clinical response
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Monitoring	☐ Improvement in the work of breathing and accessory muscles use ☐ Improvement in oxygen saturation and blood gas ☐ Frequent clinical assessment required ☐ Most achieve clinical response within first 60 -90 minutes
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Improvement in work of breathing, saturation and blood gas

☐ Continue clinical monitoring

☐ Consider feeding via NGT feeds within 4 hours of clinical response

Non-improvement in the work of breathing, saturation or blood gas

☐ Consult PICU

☐ Escalate treatment to invasive ventilation if in respiratory failure or non-invasive ventilation if not in respiratory failure

☐ Go to non-invasive ventilation algorithm

High Flow Nasal Cannula(HFNC) weaning Algorithm

Assess readiness to wean of HFNC 24 hours post initiation

Wean Fio2 if saturation is > 94% following first 2 hours of stabilization
Wean Fio2 to 40% gradually as able

improvement In respiratory distress
Use Respiratory assessment score(RAS)

Reduce Flow by 50% every 12 hours if no respiratory distress
Assess RAS 20 minutes from the flow change

Respiratory Assessment Score

Age Variable	Respiratory Rate		
	No/mild	Moderate	Severe
< 2 mos	<= 60	61 - 69	>= 70
2 mos – 1 yr	<= 50	51 - 59	>= 60
1 – 2 yr	<= 40	41 - 44	>= 45
2 – 3 yr	<= 34	35 - 39	>= 40
4 – 5 yr	<= 30	31 - 35	>= 36
6 – 12 yr	<= 26	27 - 30	>= 31
>12 yr	<= 23	24 - 27	>= 28
Intercostal Retraction	none	minimal	marked
Xiphoid Retraction	none	minimal	marked
Nasal Flaring	none	minimal	marked
Expiratory Grunt	none	audible with stethoscope	audible

	UPPER CHEST	LOWER CHEST	XIPHOID RETRACT.	NARES DILAT.	EXPIR. GRUNT
GRADE 0					
GRADE 1					
GRADE 2					

Remained free of distress

- ☐ Convert to low flow oxygen if flow rate < 1l/kg/min
- ☐ Continue on low flow oxygen and monitor for respiratory distress
- ☐ Consider transfer out if remains free of distress for 12 hours

Moderate respiratory distress
Go to previous flow and reassess weaning in 12-24 hours

Significant respiratory distress

- ☐ Stay on initial flow rates
- ☐ Assess need to escalate to NIV
- ☐ Consider slow weaning of 1 l every 4-6 hours in those with difficult to wean

Suggested low flow oxygen rates

Age	Flow Rate (L/min)
< 2 months	0.5
2 months – 2 years	1-1.5
3 – 4 years	2-3
> 4 years	No more than 4